qwkit 0.5 user guide

Installation, conventions, module reference, worked examples and reproduction of published results

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1 What the package does

`qwkit` computes the optical response of doped quantum wells from electron states to harmonic power in a waveguide, and regenerates every number and figure of the accompanying papers. It chains six calculations:

1. **Subband states** of a layered $In_{0.53}Ga_{0.47}As/In_{0.52}Al_{0.48}As$ structure under a static field (single conduction band, BenDaniel–Duke operator, Kane nonparabolicity) — `core`.

2. **Self-consistent electrostatics** of one doped period of a voltage-biased stack (periodic Schrödinger–Poisson), depolarization shifts, and the resonant second-order susceptibility $\chi^{(2)}(2\omega;\omega,\omega)$ for arbitrary populations — `sp`.

3. **LO-phonon intersubband scattering rates** (Fröhlich interaction) with detailed balance — `scattering`.
4. **TM modes** of a planar multilayer waveguide with complex permittivities — `waveguide`.
5. **Fields under alternating-polarity electrodes** and the QPM Fourier coefficient — `electrodes`.
6. **Saturated, pump-depleted SHG propagation** with multilevel rate equations and QPM patterns — `propagation`.

2 Requirements and setup

Python 3.10 or later with `numpy` and `scipy`; `matplotlib` only for the figures. From the top folder of the archive (the one containing `pyproject.toml`), install with `pip install .`, or `pip install -e .[test,plots]` for development; `import qwkit` then works from anywhere. Without installing, scripts in the archive's subfolders add the top folder to `sys.path` themselves, and for other scripts you can set `PYTHONPATH` to the top folder:

```
cd qwkit
python3 tests/run_tests.py # 25 validation tests, about 1 minute (or: pytest)

python3 docs/examples.py # the worked examples of Section 5
```

Run the test suite after any change to the package and before regenerating figures.

3 Units and conventions

Quantity	Unit inside the package
Position z, layer thicknesses, dipoles z_{ij}	Å (`core`, `sp`, `scattering`)
Energies, potentials, linewidth Γ (HWHM), photon energy $\hbar\omega$	eV
Applied field	kV/cm (positive F lowers the potential towards $+z$)
Sheet density per period n_s	cm^{-2}; volume density N in cm^{-3}
$\chi^{(2)}$ from `core`/`sp`	pm/V (complex); `Medium.chi` returns m/V
Waveguide lengths, wavelengths	μm; modal loss in cm^{-1}; overlap $\lvert\int f_1^2 f_2^*\rvert^2$ in $\mu\mathrm{m}^{-1}$
Electrode geometry (`electrodes`)	m; field in V/m
Intensities in `Medium.populations`	$\mathrm{W/m}^2$ (10^{10} $\mathrm{W/m}^2$ = 1 $\mathrm{MW/cm}^2$)
Pump power	W; scattering rates in s^{-1}

A structure is a list of `(thickness_Å, kind)` tuples, with `kind` a key of `core.MAT` (`"well"` = InGaAs, `"barrier"` = InAlAs); register other materials with `qwkit.add_material(name, V, m, Eg, eps)`. The self-consistent routines pad the structure with half of the spacer on each side (default spacer 200 Å of InAlAs) and place the donors as sheets at the period boundary. Photon energy for a 10 μm pump: `HW = 1.23984/10`.

4 Module reference

4.1 `core` — states and susceptibility

Function	Purpose and returns
`build(layers, pad=100, dz=0.5)`	Cell-centred grid `z` and material arrays `V0, m, Eg`; `active` = (start, end) of the structure.
`potential(z, V0, active, F_kVcm)`	Adds a uniform field, zero of tilt at the structure centre.
`solve_states(z, V, m_edge, Eg, nstates=6, nonparabolic=True, E_guess=None)`	Energies (eV) and normalized wavefunctions (columns, $\sum \lvert\psi\rvert^2 dz = 1$). Index tracking keeps the k-th state the k-th eigenvalue of $H(E)$; pass `E_guess` to warm-start.
`localized(z, psi, active, thresh=0.8)`	Indices of states with $\geq$80% probability in the active region.
`dipoles(z, psi)`	Matrix z_{ij} (Å).
`populations(E, ns_cm2, T)`	Fermi–Dirac subband fractions for a parabolic 2D density of states.
`chi2_general(E, Z, rho, hw, gamma, N_cm3)`	Density-matrix $\chi^{(2)}$ (pm/V) for arbitrary populations; translation invariant.
`chi2_shg`, `alpha_w`, `analyze`, `analyze_T`	Ground-state-only $\chi^{(2)}$, linear absorption, and convenience wrappers (fixed potential).
`wkb_escape_time`, `escape_broadening`	WKB tunnelling estimates for biased levels.

4.2 `sp` — self-consistent electrostatics and depolarization

Function	Purpose and returns
`chi2_sc(layers, F_kVcm, hw, ns_cm2=5e11, T=300, gamma=0.010, spacer=200, dz=0.5, doping="spacer", depol=True, sc=True, VH0=None)`	Main entry point. Returns a dict with `chi` (pm/V), `E`, `Et` (depolarization-shifted transition matrix), `Z`, `rho`, `N` (cm^{-3}) and `r` (the raw self-consistent solution, including `r["VH"]` for warm starts). `sc=False` or `depol=False` switch the physics off for model-ladder studies; `doping` is `"spacer"`, `"wells"` or `"centre"`.
`self_consistent(...)`	Periodic Hartree iteration at fixed mean field; returns `z, V, VH, E, psi, ...`.
`prepare(r, nkeep=6)`	Localized, Löwdin-orthonormalized states, dipoles about the structure centre, populations: `E, psi, Z, rho`.
`depol_energies`, `chi2_depol`	Ando depolarization shifts ($\varepsilon = 13.5$) and $\chi^{(2)}$ with shifted energies.
`state_set(o)`	Converts a `chi2_sc` result into the state-set dict used by `propagation.Medium`.

4.3 `scattering` — LO-phonon relaxation

`lo_rate(z, psi_i, psi_f, dE, T=300, hwlo=0.0337, eps_inf=11.6, eps_s=13.9, m_eff=0.043)`	Downward rate (s^{-1}) from the bottom of subband i to f, emission plus absorption, Eq. (7) of the paper.
`rate_matrix(z, psi, E, T=300, **kw)`	Matrix R_{lm} (s^{-1}, from l to m); upward rates by detailed balance.
`form_integral(z, phi, Q)`	$I(Q)$ in O(N) operations.

4.4 `waveguide` — planar TM modes

Layers are dicts `dict(material=..., t_um=..., N=..., mu=...)` with `material` one of `"Au"`, `"InP"`, `"InGaAs"`, `"InAlAs"`, `"stack"` (needs `frac_well`) or `"air"`, listed from the top. `tm_modes(layers, lam_um, dz_um=0.01, nmodes=3)` returns mode dicts with `neff` (complex), `alpha_cm`, and the profiles `z`, `Ez`. `overlap(mode, z0, z1)` gives the confinement factor of $|E_z|^2$ in $[z_0, z_1]$ (μm from the top), and `shg_overlap(m1, m2, z0, z1)` the nonlinear overlap $|\int f_1^2 f_2^* dz|^2$ ($\mu\mathrm{m}^{-1}$). Mid-infrared indices of the ternaries are assumed (3.43 and 3.20 at 10 μm with InP dispersion) and set the coherence length only approximately.

4.5 `electrodes` — bias field and QPM coefficient

`field_map(Lam, d, V, gap, y, nx, nmax)` returns x and $E_y(x)$ at height y above the ground plane for segments $\pm V$ of period `Lam` separated by gaps `gap` (all in m), uniform permittivity. `qpm_coefficient(x, chi_x, Lam)` returns $|\Lambda^{-1}\int \chi(x) e^{-2\pi i x/\Lambda} dx|$; it equals $(2/\pi)\chi_0$ for an ideal $\pm\chi_0$ square wave.

4.6 `propagation` — saturated SHG

`Medium(st, hw, gamma=0.010, tau_ps=1.0, nref=3.3, R=None)` builds a saturable medium from a state set `st = dict(E, Z, Et, rho, N)` (lists, as stored in `studies/n1_paper/tables.json`). Without `R` it uses a single relaxation time; with `R` (from `scattering.rate_matrix`) it uses level-resolved relaxation. Methods: `populations(I1, I2)`, `chi(rho)` (m/V), `alpha(rho)` (1/m at ω and 2ω).

`propagate(med, wg, P1, scheme, eta_e=1.0, L_max_um=5000, steps_per_lcoh=32)` integrates the coupled-mode equations with RK4. `wg` is a dict with `n1`, `n2`, `G1`, `G2`, `O_um`, `abg1_cm`, `abg2_cm`, `t_stack_um`, `w_um`, `Lcoh_um`, `lam1_um`. `scheme` is `"none"`, `"perfect"` ($\Delta k = 0$), `"sign"` ($\pm$`eta_e`, the electrode factor) or `"onoff"`. It returns `x_um`, `P1`, `P2`, the peak efficiency `eta` $= \max P_2/P_1(0)$ and its position `L_opt_um`.

`propagate_segments(segs, wg, P1, L_max_um)` handles media that differ from segment to segment: `segs` is a list of `(Medium, Lcoh_um, s)` cycled along the guide, each segment one coherence length long with its own $\chi^{(2)}$, loss and phase mismatch (used for realistic Stark on/off QPM).

5 Worked examples

The file `docs/examples.py` runs the following steps; its output is in `docs/examples_output.txt`.

```
import numpy as np
from qwkit import sp, scattering, waveguide, electrodes, propagation
HW = 1.23984 / 10
```

```
TW = [(58, "well"), (14, "barrier"), (34, "well"), (14, "barrier"), (58, "
    well")]
AD = [(64, "well"), (10, "barrier"), (29, "well")]

tw = sp.chi2_sc(TW, 55.0, HW, ns_cm2=5e11) # 1. self-consistent chi(2)
ad = sp.chi2_sc(AD, 0.0, HW, ns_cm2=5e11)
m = sp.chi2_sc(TW, -55.0, HW, ns_cm2=5e11) # 2. odd in F: tw["chi"] + m["chi
    "] ~ 0

E, psi, Z, rho = sp.prepare(tw["r"]) # 3. LO-phonon rates
R = scattering.rate_matrix(tw["r"]["z"], psi, E)

med = propagation.Medium(sp.state_set(tw), HW, R=R) # 4. saturable medium
rho_I = med.populations(1e10, 0.0) # 1 MW/cm^2 local pump intensity
print(abs(med.chi(rho_I)) * 1e9, "nm/V")

wg = dict(n1=3.231, n2=3.338, G1=0.395, G2=0.514, O_um=0.0531, abg1_cm=3.25,
        abg2_cm=0.16, t_stack_um=1.512, w_um=15.0, Lcoh_um=23.4, lam1_um
           =10.0)
res = propagation.propagate(med, wg, 1.26, "sign", eta_e=0.92, L_max_um
    =500.0)
print(res["eta"] * 100, "%␣at", res["L_opt_um"], "um") # 7. saturated SHG
```

Expected output (abridged):

```
1. |chi2| triple well 63.8 nm/V, ADQW 120.7 nm/V
2. chi2(+F) + chi2(-F) = 1.40e-06 pm/V (should be ~0)
3. triple-well lifetimes: 2->1 2.62 ps, 3->2 0.79 ps, 3->1 1.96 ps
4. I = 1 MW/cm2: |chi2| 35.1 nm/V, alpha(w) 490 1/cm
5. n_eff 3.231 / 3.338, Gamma 0.39 / 0.52, loss 3.30 / 0.16 1/cm, Lcoh 23.5
    um
6. plateau field 54.6 kV/cm; QPM coefficient of sign(F) = 0.637 (ideal 2/pi
    = 0.637)
7. 1.26 W pump: peak SH conversion 0.121% at 63 um
```

Example 5 uses a 0.02 μm grid for speed; the paper uses 0.01 μm (confinement 0.51 at 5 μm, $L_{\mathrm{coh}} = 23.4$ μm).

Designing a new structure. Replace `TW` by your layer list, scan the field with a warm start, and compare with the ADQW at the same n_s:

```
VH = None
for F in range(0, 105, 5):
   o = sp.chi2_sc(my_layers, float(F), HW, ns_cm2=5e11, VH0=VH); VH = o["r"
      ]["VH"]
   print(F, abs(o["chi"]) / abs(ad["chi"]))
```

New materials. Add an entry to `core.MAT` (`V` offset in eV relative to InGaAs, band-edge mass `m`, gap `Eg`) and to `sp.EPS_R`, as `studies/n1_paper/stepwell.py` does for InGaAlAs quaternaries. For a different well material, also change the LO-phonon parameters passed to `scattering.rate_matrix`.

6 Reproducing the papers

Software paper. `python3 examples/validation/software_figures.py` regenerates both figures of the software paper and `paper_software/timing.json`; build the paper with `cd paper_software; pdflatex qwkit_software; bibtex qwkit_software; pdflatex qwkit_software; pdflatex qwkit_software`.

Triple-well QPM study. From `studies/n1_paper/`, with the top folder on `PYTHONPATH`, run the scripts in this order (total about 30 minutes on two cores). Each writes a JSON file read by later steps and by `paper_figures.py`.

Script	Output	Used for
`tables.py`	`tables.json`	$\chi^{(2)}(F)$; state sets for all n_s
`curves_sc.py`	`curves_sc.json`	Fig. 2
`stepwell.py`	`stepwell.json`	stepped-well benchmark (Sec. 4.1)
`waveguide_design.py`	`waveguide.json`	Sec. 4.4, Fig. 4a
`electrode_study.py`	`electrodes.json`	electrode factor, Fig. 4b
`linear_grating.py`	`linear_grating.json`	residual linear grating (Sec. 4.4)
`rates_study.py`	`rates.json`	LO-phonon rates, saturation intensities (Sec. 4.5)
`propagation_lo.py`	`propagation_lo.json`	Table 3, Fig. 5
`stark_realistic.py`	`stark_realistic.json`	realistic Stark QPM
`sensitivity.py`	`sensitivity.json`	Table 4
`propagation_study.py`	`propagation.json`	single-τ comparison
`adqw_sign.py`	`propagation.json`	single-τ, growth-reversed ADQW
`summary_lo.py`	`summary_lo.json`	quoted ratios, crossovers, heating
`paper_figures.py`	`paper/figures/*.pdf`	Figs. 1–5

Table 2 and Fig. 3 come from `studies/n1/ladder.py 58 14 34 50 64 10 29` and `studies/n1/errbudget.py 58 14 34 64 10 29`. Build the manuscript with `cd paper; pdflatex main; bibtex main; pdflatex main; pdflatex main`.

7 Validation

`tests/run_tests.py` (or `pytest`) runs 25 tests (Table 2 of the software paper): analytic infinite-well limits, the three-level limit of the book formula, exact symmetry ($\chi^{(2)}(-F) = -\chi^{(2)}(F)$ to 10^{-6}), translation invariance, parity with nonparabolicity, charged-sheet electrostatics, neutrality and convergence, depolarization limits, the symmetric slab waveguide, parallel-plate electrodes, QPM coefficients, the undepleted SHG limit, power conservation, two-level saturation, the LO-phonon form factor (brute force and short-range limit), a published LO-phonon benchmark, detailed balance, level-resolved rate equations, segmented propagation, the analytic finite well with BenDaniel–Duke matching (with second-order grid convergence), and the phase-mismatched sinc^2 limit. `examples/validation/software_figures.py` turns four of these checks and an end-to-end example into the figures of the software paper. Add a test for every new physical quantity, checked against an analytic limit or a published number.

8 Limitations

Single conduction band with literature parameters; Lorentzian lineshapes; first-order depolarization (no exciton-like or local-field corrections); LO-phonon scattering for electrons at the subband minima with bulk phonons (no interface-roughness, alloy or electron–electron scattering); parabolic 2D density of states

for populations; steady-state rate equations (valid for pulses much longer than 1 ps); planar waveguide; uniform-permittivity electrode model; assumed mid-infrared alloy indices; no thermal or transport model. These limitations are discussed in Sec. 5 of the paper.

9 Troubleshooting

- *`ModuleNotFoundError: qwkit`*: run from the top folder or set `PYTHONPATH` to it.
- *Fewer states than expected*: `prepare` keeps only states with 80% of their probability in the active region; at high fields upper levels become quasi-bound and drop out (this is physical).
- *Slow field scans*: pass the previous `r["VH"]` as `VH0` to warm-start the Hartree iteration.
- *Spurious $\chi^{(2)}$ at zero field in a symmetric structure*: the structure or doping is not mirror-symmetric, or the grid step does not divide the layer thicknesses evenly; keep `dz=0.5` Å with integer-Å layers.